Model	Macro-F1		%Failed	
	Infomal Def.	Formal Def.	Infomal Def.	Formal Def.
GPT-4	48.38	**49.78**	0.67	0.83
GPT-3.5	31.27	27.10	2.50	4.00
Qwen2.5-14Bf	31.11	**34.64**	0.88	0.70
Qwen2.5-7Bf	30.62	29.43	5.00	2.20
Llama3-8Bf	24.43	**25.71**	12.90	12.10
Mistral-7Bf	22.23	18.37	3.50	3.60
Llama2-13Bf	22.09	21.30	27.70	8.10

Table 5: Informal vs. formal fallacy definitions in the single-round prompting scheme on MAFALDA.

Model	Shot	LOGIC	ELECDEB	PROPAGANDA
GPT-4	One	**54.48**	**44.36**	33.18
	Two	**55.14**	**44.64**	31.57
	Zero°	48.45	42.26	34.80
	Zero*	50.54^{GFA}	42.26°	$\mathbf{35.37}^{\text{CoT}}$

Table 6: Few-shot vs. zero-shot results of GPT-4. °: Zero-shot single-round results without definitions. *: Best zero-shot results. See Table 23 for other LLMs.

of news and political speeches are delivered in formal language or specific utterance styles and rely on sufficient contexts for comprehension. It could be hard to infer about the fallacy type if the understanding of the content and context is challenged by the truncation of discourses. Besides, the fallacious examples in the three in-domain datasets, though may share the same fallacy names with other datasets, could have nuanced differences in definitions that deviate from the common ones (Alhindi et al., 2023a) and refer to specific language use cases tied to the domain context.

4.2 Single-Round vs. Multi-Round

As shown in Table 2, our multi-round prompting schemes are effective in further improving LLMs' zero-shot performances compared to the basic single-round prompts. In detail (See Table 24), they improve on 83.7% of the classification experiments compared to the basic single-round prompting schemes, with an average improvement across the datasets of 1.63, 2.46, 4.95, 3.67, 7.54, 4.02 and 1.27 points for GPT-4, GPT-3.5, Qwen2.5-Instruct (14B), Qwen2.5-Instruct (7B), Llama3-Chat (8B), Mistral-Instruct (7B) and Llama2-Chat (13B), respectively, indicating the effectiveness of our proposed multi-round prompts, especially for capable small LLMs such as Llama3 and Qwen2.5. Multi-round prompted GPT-4 is also further improved on most tasks compared to its best single-round prompting performance and achieves new state-of-the-art scores on ARGOTARIO, MAFALDA and COVID-19.

4.3 Multi-Round Prompting Schemes Analysis

We present the overall average Macro-F1 ranking of each multi-round prompting scheme in Table 3. It shows that three of our proposed multi-round prompting schemes can be superior to the basic single-round prompts. General Fallacy Analysis with or without the warm-up round (GFA-W and GFA) ranks within the top three best-performed multi-round schemes, indicating that explicit fallacy analysis can benefit LLMs by eliciting their internalized fallacy knowledge to guide the classification process. However, all the proposed multi-round prompting schemes that require LLMs to first perform tasks that indirectly address the focal classification problem could be prone to failed classification *i.e.*, predicting a fallacy type that is outside the given fallacy label space or predicting as non-fallacious. This is predicable since it is possible that some data examples do commit more than one type of fallacy or can be interpreted as certain out-of-scope fallacies. Having LLMs to make analysis first without limiting them within a specific label space increases the probability that LLMs bias to their first judgements. An additional warm-up round that allows LLMs to first reflect on the content and context of input discourses can not only improve the overall classification performance but also help alleviate the failure rate (12.27% vs. 15.09%). Zero-shot CoT generally ranks as the second-best multi-round prompting scheme, suggesting that CoT's general contribution to improved reasoning in LLMs is also transferable to fallacy classification tasks. Besides, Zero-shot CoT has significant advantages over GFA(-W) if taking the average failure rate into account, which makes Zero-shot CoT the potentially most effective and robust multi-round prompting scheme in practice.

Although on average Definition Generation slightly underperforms the single-round With Definitions, it is still better than prompting LLMs with only the label space. Human crafted fallacy definitions that could be more detailed and relevant to the focused context, if available, are usually more reliable than LLMs' self-generated definitions. It is notable that both Zero-shot CoT and Definition Generation have lower failure rates, probably because they have fewer chat rounds and can

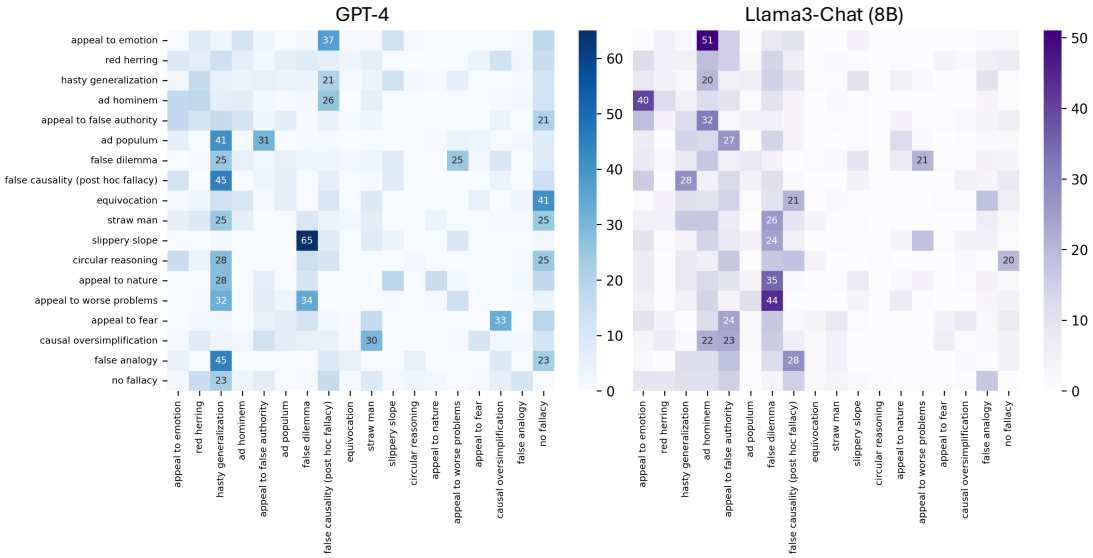

Figure 3: Misclassification confusion matrix of common fallacy types given by GPT-4 and Llama3-Chat (8B). Rows are the ground truth fallacy types and columns are predicted fallacy types. Cell values represent the percentages of row fallacy types that are misclassified as column fallacy types.

immediately address the given label space at the first place, confining LLMs to more restricted contexts that are less likely to be directed to unlimited or irrelevant association. In particular, Definition Generation has the lowest failure rate, indicating that LLMs could be more robust in generating instruction-specified labels when relying on their self-generated contents. In contrast, the three-round Premises & Conclusion that explicitly applies the formal definition of the term "fallacy" is not always a robust choice to improve the fallacy classification performance of LLMs. It turns out that the formal academic conception cannot be well translated to eliciting LLMs' knowledge and abilities for distinguishing among different fallacy types but mislead LLMs to a deviated focus on checking "whether the premises entail the conclusion" rather than recognizing the specific fallacy type. (See Appendix D.1)

We present the top two best performed prompting schemes for different classification scenarios with respect to data domains and model types in Table 4. This summarization may serve as a general guidance for fallacy classification in practice. As suggested by the results, when applying LLMs to open-domain fallacy classification tasks in a zero-shot setting, the inherent pre-trained fallacy knowledge of advanced GPT models could be effective when explicitly prompted for self-generated fallacy definitions or directly referring to human crafted defintions. For smaller LLMs, they can be prompted to reflect on the input context before giving a general fallacy analysis to better elicit and leverage their internalized relevant knowledge. When tackling domain-specific fallacy classification tasks, to address the nuances in fallacy definitions for in-domain characteristics, it is more effective to require GPT models to directly refer to the manually crafted definitions, while for small LLMs, they can still benefit from Zero-shot CoT prompting. See Table 21 and Table 22 for more detailed rankings and the comparison between best zero-shot results and the guidance-suggested results in Table 25.

4.4 Informal vs. Formal Definitions

We substitute the informal fallacy definitions (Table 27) with formal fallacy definitions (See Table 28) of MAFALDA summarized by Helwe et al. (2023) in the single-round prompting scheme. As shown in Table 5, we can only observe a limited improvement of performance on more capable LLMs of GPT-4, Qwen2.5-Instruct (14B) and Llama3-Chat (8B) when providing them with formal definitions of fallacies. The results of the other LLMs remain comparable with or are slightly worse than that with informal definitions. The failure rates of the two prompting schemes are not significantly different. While formal definitions that incorporate some symbolic representations could be less ambiguous, they are not necessarily easier for LLMs to comprehend and relate to the exact occurrences of fallacies in actual discourses.

4.5 Few-shot vs. Zero-shot

To verify whether LLMs' weak performance on some datasets (*i.e.*, LOGIC, ELECDEB and PROPAGANDA) could have been resulted from the challenging fully zero-shot prompt setting, we conduct N-way few-shot experiments that randomly present LLMs with 1-shot or 2-shot examples of each fallacy classes from the holdout data splits. Experiment results in Table 6 show that only GPT-4's performance can be further improved on LOGIC and ELECDEB. The rest of few-shot results by other LLMs all fail to surpass the previous best-performed zero-shot prompting results, suggesting the limitation of LLMs' in-context learning abilities in handling hard and in-domain fallacy classification tasks.

4.6 Error Analysis

To view how LLMs perform on different fallacy types, we present the confusion matrices of GPT-4 and Llama3-Chat (8B) on each dataset in Appendix D.3. To further analyze classification errors, we aggregate the experimental results of GPT-4 and Llama3 on each dataset under their top three best performed zero-shot prompting schemes and select a total of 18 common fallacy types (including *No Fallacy*) that occur in more than two datasets to present the confusion matrix in misclassification percentage as shown in Figure 3. It is notable that GPT-4 is generally more inclined to classify fallacy examples to be non-fallacious compared to Llama3. As revealed in Figure 7 (a) and Figure 9 (e), GPT-4 can better recall actual *No Fallacy* examples but severely over predict most examples in COVID-19 to be non-fallacious as shown in Figure 8 (b). In addition, some fallacy types frequently confused by LLMs reflect the inherent limitation in the natural language definitions of informal fallacies, which may overlap with each other with inevitable ambiguity. For example, GPT-4 tends to classify many fallacy types as *Hasty Generalization*. We find that GPT-4's self-generated definition for *Hasty Generalization* that "*A conclusion based on insufficient or biased evidence; rushing to a conclusion before you have all relevant facts*." can be applied to interpret many fallacies *e.g.*, *Ad Populum*, *Post Hoc Fallacy* and *False Analogy*. We find similar explanations towards both GPT-4's and Llama3's misclassifations of *Appeal to Worse Problems*, *Slippery Slope* as *False Dilemma* and *Ad Populum* as *Appeal to False Authority*. As for Llama3, we find that it tends to overpredict the fallacy types of *False Dilemma*, *Ad Hominem* and *Appeal to False Authority*. Interestingly, as shown in the corpus of U.S. presidential debate ELECDEB in Figure 8 (d), Llama3 predicts more *Appeal to Emotion*, *Ad Hominem* and *Appeal to False Authority*, which are more basic and superficial fallacy types, while GPT-4 predicts more *Slippery Slope* and *Post Hoc Fallacy* that are fallacy conceptions more related to causal reasoning. This further suggests that the task of fallacy classification requires advanced reasoning abilities. We further analyze the potential reason for the relatively poor performance of GPT-based models under the zero-shot multi-round prompting scheme of Premises & Conclusion in Appendix D.1.

5 Conclusion

In this paper, we explore LLMs' performance on zero-shot fallacy classification. We propose both single-round and multi-round prompting schemes to fully elicit LLMs' fallacy classification ability. Through extensive experiments on benchmark datasets, we demonstrate that LLMs with a zero-shot single-round prompting scheme can outperform or achieve highly comparable performances with SOTA full-shot fine-tuned T5 baselines on some open-domain benchmark datasets and can generally achieve sub-optimal performances on hard domain-specific datasets. Besides, our multi-round prompting schemes can effectively enhance the performances, especially for small LLMs. The performance gaps between zero-shot prompted LLMs and full-shot fine-tuned baselines are acceptable, indicating LLMs' potential for further improvements for this task. Our detailed analysis moreover highlights the future research for zero-shot fallacy classification.

Limitations

We consider the following limitations for future work. First, we only focus on single-label fallacy classification. For instance, we only use the dominant types in the dataset MAFALDA as the ground-truth labels. We may later explore the more complex multi-label fallacy classification task. Second, since we only conduct few-shot experiments on instruction-tuned LLMs that are optimized for chat-based applications, it is yet to be explored whether vanilla LLMs could better leverage their in-context learning abilities for the few-shot classification.